Characteristics of the proposed dams - Blocks 2, 4 and 6

The three tables below list the proposed dams of the Renardet blocks in one layout: dam type, height above the wadi bed, crest length, full supply level and reservoir capacity at full supply level. Height above wadi is the crest elevation minus the lowest ground along the dam axis; crest length is the length of the dam axis, closure sections included, dikes excluded. The Block 6 table is the one issued with the Block 6 masterplan; the Block 4 and Block 2 tables are built on the same definitions from the current dam axes, levels and reservoir capacities.

Table 1 - Characteristics of the proposed dams in Block 2

Dam Name	Dam Type	Height above wadi (m)	Crest Length (m)	FSL (m a.s.l.)	Reservoir Capacity (MCM)
Bidah 03	Not decided	15	1,335	94	2.2
Faydh 01	Not decided	49	920	219	7.4

Dam types in Block 2 are not yet decided; the two dams are second-priority sites of the approved list of priority dams (November 2024). Crest levels 99 m (Bidah 03) and 224 m (Faydh 01) a.s.l.

Table 2 - Characteristics of the proposed dams in Block 4

Dam Name	Dam Type	Height above wadi (m)	Crest Length (m)	FSL (m a.s.l.)	Reservoir Capacity (MCM)
Hawasinah 02	GFRD	20	3,160	144	11.0
Hawasinah 11	GFRD+RCC	40	960	251	34.0
Hawasinah UP**	GFRD	22	9,690	60	81.0
Shafan 02	GFRD+RCC	39	2,440	235	43.0
Shafan 08	GFRD	21	4,270	168	15.0
Sarami 03	GFRD	46	1,530	237	45.0
Sarami 05	RCC	26	2,550	184	15.0
Sakhin 04	GFRD	26	2,810	122	24.0
Sakhin 08	Not decided	18	2,070	139	8.0

*** Upgradation of the existing dam. GFRD+RCC: embankment with an RCC section across the wadi; the type of Sakhin 08 is not yet decided. Sarami 03 and Sakhin 08 carry an additional dike of about 480 m and 940 m.*

Table 3 - Characteristics of the proposed dams in Block 6

Dam Name	Dam Type	Height above wadi (m)	Crest Length (m)	FSL (m a.s.l.)	Reservoir Capacity (MCM)
Fulaij 05	FSHD	28	660	125	18.5
Halban EX*	GFRD	9	3,040	64	3.7
Taww UP**	GFRD	17	7,380	69	45.2
Maawil 04**	RCC	57	270	413	11.8
Maawil 07	RCC	48	170	399	3.6
Maawil UP**	GFRD	18	10,400	68	74.8
Abyad 01	RCC	94	355	310	125.0
Bani Kharus UP**	GFRD	20	11,760	75	116.0
Ais 10	GFRD	19	4,520	207	13.4

** Rehabilitation of the existing dam. ** Upgradation of the existing dam.*

Dam types: FSHD - faced symmetrical hardfill dam; GFRD - geomembrane-faced rockfill (granular fill) dam; RCC - roller-compacted concrete gravity dam.

Basis and sources (internal - not for the report)

Block 6: the dam team's details sheet, 07-Structures\03-Dam Design\01-Block 6 Dams\Block 6 Dams Details.xlsx, reproduced without change; it is Table 34 of the Block 6 masterplan report and the table still sitting in the Block 4 masterplan draft. Their heights and crest lengths come from the axis profiles on the 2 m DEM (Characteristics*.dxf): crest minus the lowest profile point, and the axis length.

Block 4: FSL and reservoir capacity from the dam team's Block 4 Dams Details.xlsx (identical to the hydraulic copy in 04-Hydraulic\00 Data\Dams\Block 04); dam type from their Chracteristics\Block 4 Dams.xlsx; height and crest length derived here from their Chracteristics*.dxf profiles with the Block 6 definition (the Block 6 values re-derived the same way agree within 1 m and 1 %, Taww UP excepted: 20.1 m on the profile against 17 m in the sheet). The Block 4 hydrology report (R4) states the height as crest minus the reservoir bed of the EAV curve, which differs from these by at most 1 m.

Block 2: axes and reservoir polygons of January 2025 (04-Hydraulic\00 Data\SHP\1st_2nd_Priority_DAM_AXIS_ART_REN 08.01.2025), profiled on the 2 m DEM (13-Study and Design\01-FS and PD\00-General\01-Topography\NSA_Batinah_2m.tif). The axis ends sit at the crest level and the reservoir polygon boundary at the FSL, which confirms the levels of the 2023 feasibility ranking (07-Structures\02-Potential Dam Sites\02-Feasibility\02-Ranking Matrix\Dam site Characteristics.xlsx: FSL 94 / crest 99 and FSL 219 / crest 224). Capacity is the volume below FSL inside the reservoir polygon. No dam design exists yet: the folders 07-Structures\03-Dam Design\03-Block 2 Dams, Bidah 03 and Faydh 01 are empty, the Block 02 HMS dam model has sinks at the two sites and no reservoirs, and the March 2025 feasibility slides for Bidah and Faydh carry only the general sheet. The dam team's tentative cost sheet (07-Structures\Ad-hoc Reports\03-Tentative Costs of Dams) rounds them to 25 m / 1,300 m (listed there as Bidah 01) and 50 m / 900 m, at 10.3 and 19.7 million OMR.

Per-dam derivation notes:

- Bidah 03: crest 99; 2 m DEM along the January 2025 axis (lowest ground 83.8); the 2023 feasibility ranking gave 16.8 m / 1361 m / 2.16 MCM.
- Faydh 01: crest 224; 2 m DEM along the January 2025 axis (lowest ground 175.3; 790 m of the 919 m axis lies below crest level); the 2023 feasibility ranking gave 53.1 m / 792 m / 8.0 MCM.
- Hawasinah 02: profile minimum 129.4 against crest 149.
- Hawasinah 11: main 755 m plus left closure 202 m; profile minimum 215.8 against crest 256.
- Hawasinah UP: profile minimum 43.2 against crest 65.
- Shafan 02: main 1763 m plus left 591 m and middle 89 m; profile minimum 201.5 against crest 240.
- Shafan 08: main 3235 m plus right 1032 m; profile minimum 152.5 against crest 173.
- Sarami 03: main 731 m plus three right closures 797 m; a 480 m dike besides; profile minimum 196.3 against crest 242.
- Sarami 05: profile minimum 163.4 against crest 189.
- Sakhin 04: profile minimum 101.5 against crest 127.
- Sakhin 08: main 1082 m plus two left closures 992 m; a 940 m dike besides; profile minimum 126.0 against crest 144.